

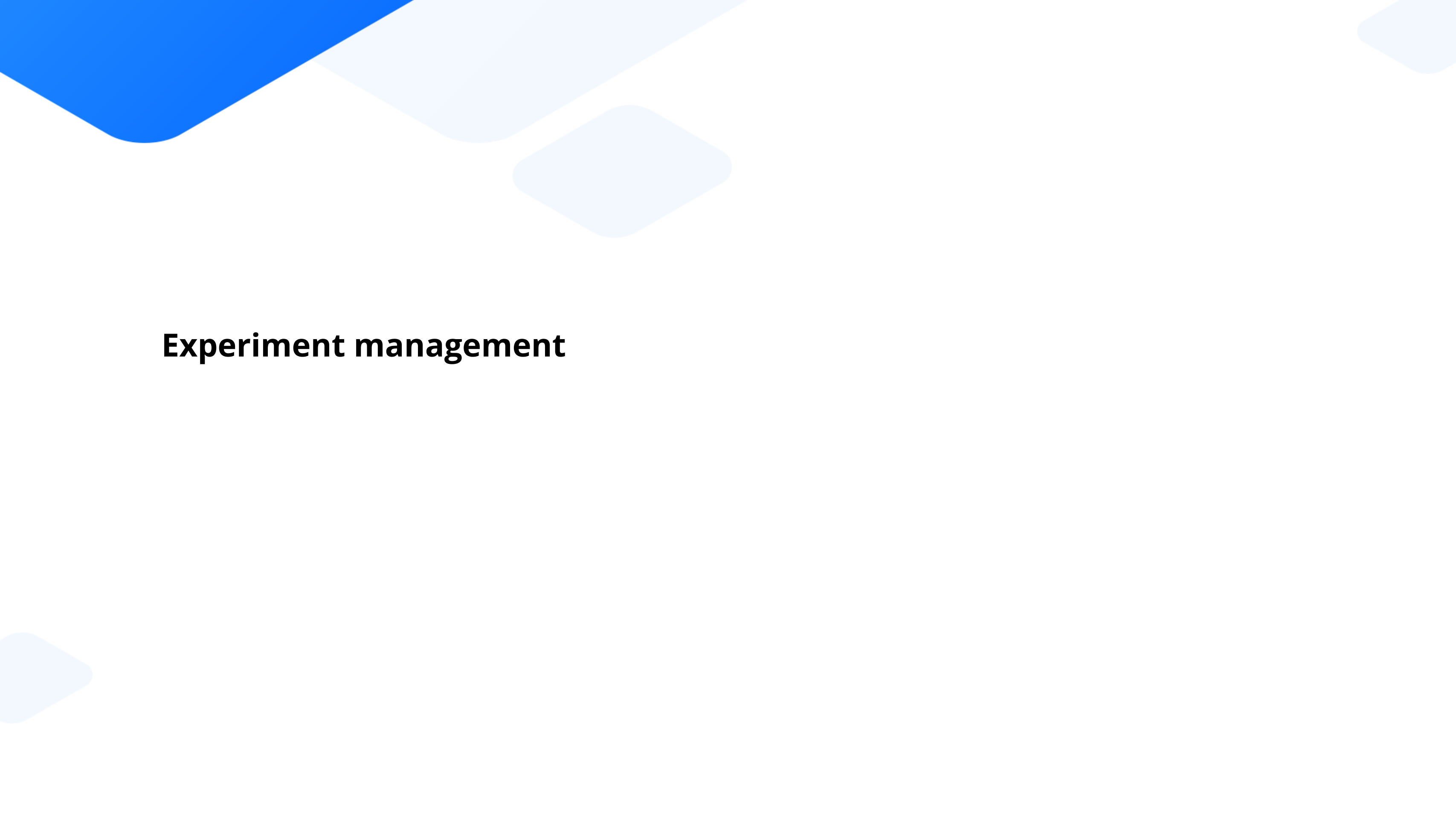
M06 - Experiment Tracking and Model Registry



Topic

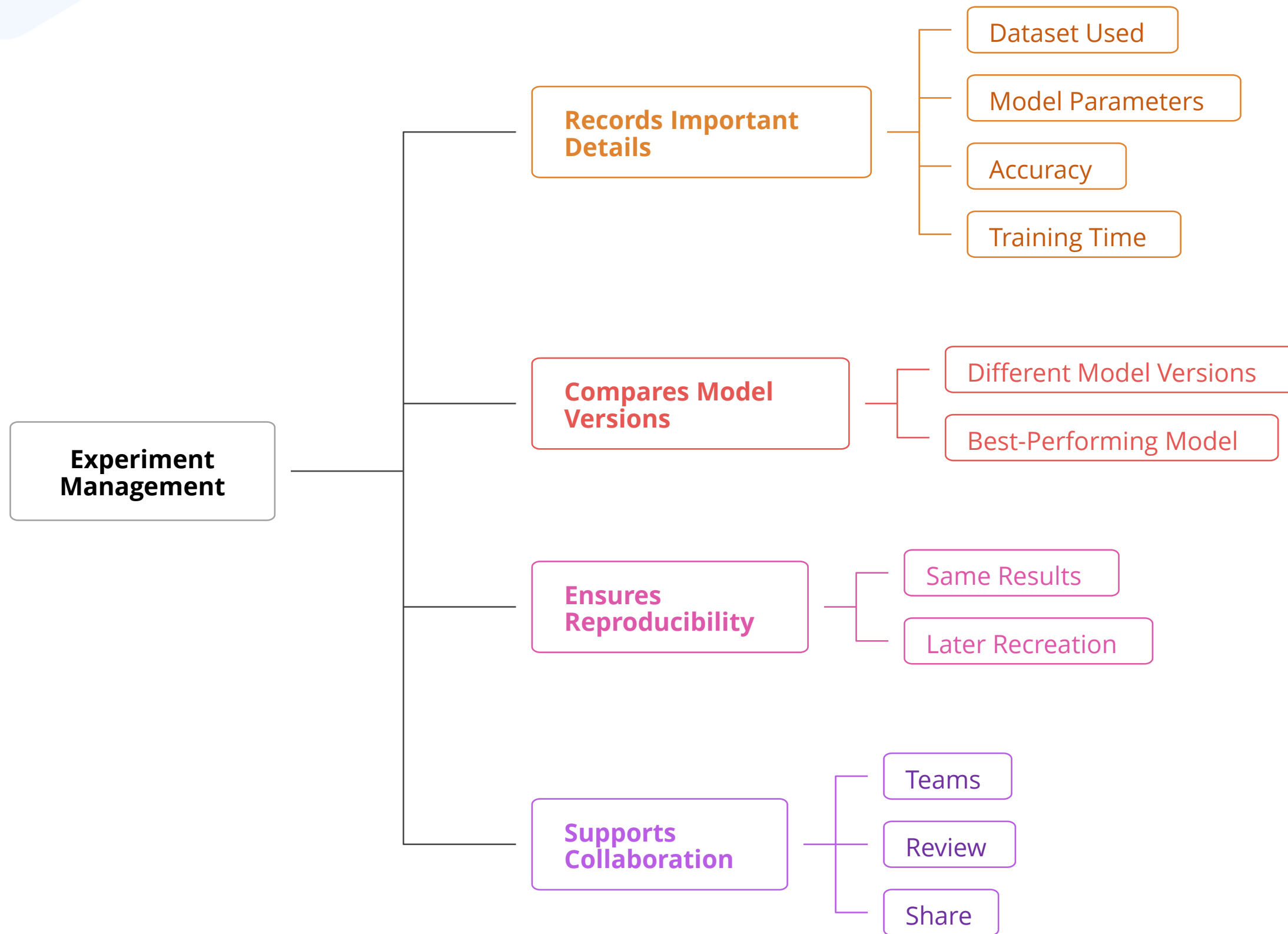
- **Experiment management**
- **Tracking metrics and parameters**
- **Model versioning**
- **Model lifecycle states**
- **Promotion workflows**
- **Registry integration with CI/CD**





Experiment management

Experiment Management in AI Model Training

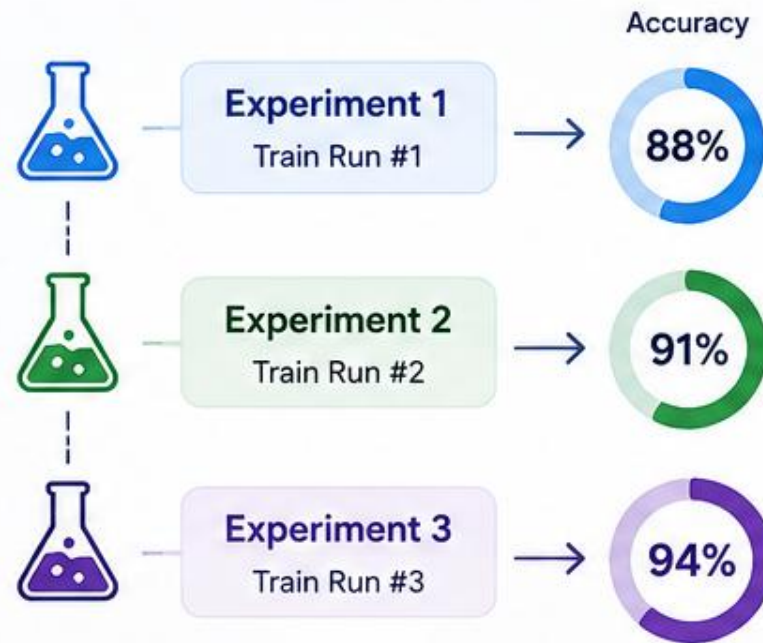


Common Experiment Management Tools

Tool	Purpose	Recommendation
 MLflow	Track experiments, parameters, metrics, and models	★★★★★ Best for your course
 Azure Machine Learning	Enterprise experiment tracking and model management	★★★★
 Amazon SageMaker Experiments	AWS experiment tracking	★★★★
 Weights & Biases (W&B)	AI experiment tracking and visualization	★★★★
 Neptune.ai	Experiment metadata and collaboration	★★★
 Comet ML	Experiment tracking and comparison	★★★

VOIS EXAMPLE – EXPERIMENT MANAGEMENT WITH MLflow

1 QA Team Trains the AI Model Three Times



2 MLflow Records Every Detail

mlflow

Model Version	v1, v2, v3
Training Date	2026-07-30
Dataset Version	ds_v1.0
Parameters	lr, max_depth, n_estimators...
Accuracy	88%, 91%, 94%
Training Time	15m, 17m, 14m

3 QA Lead Opens MLflow UI and Selects the Best Model



Best Performing Experiment

Experiment 3 **94%**

Selected for Deployment

MLflow UI – Compare Experiments

mlflow

Experiments

Models

Artifacts

Experiment: VOIS – Regression Test Prioritization

Filter

Compare

Run Name	Model Version	Dataset Version	Parameters (lr, max_depth, n_estimators)	Accuracy	Training Time	Status
Experiment 1	v1	ds_v1.0	0.01, 6, 100	88%	15m	Completed
Experiment 2	v2	ds_v1.0	0.01, 8, 150	91%	17m	Completed
Experiment 3	v3	ds_v1.0	0.01, 10, 200	94%	14m	Completed

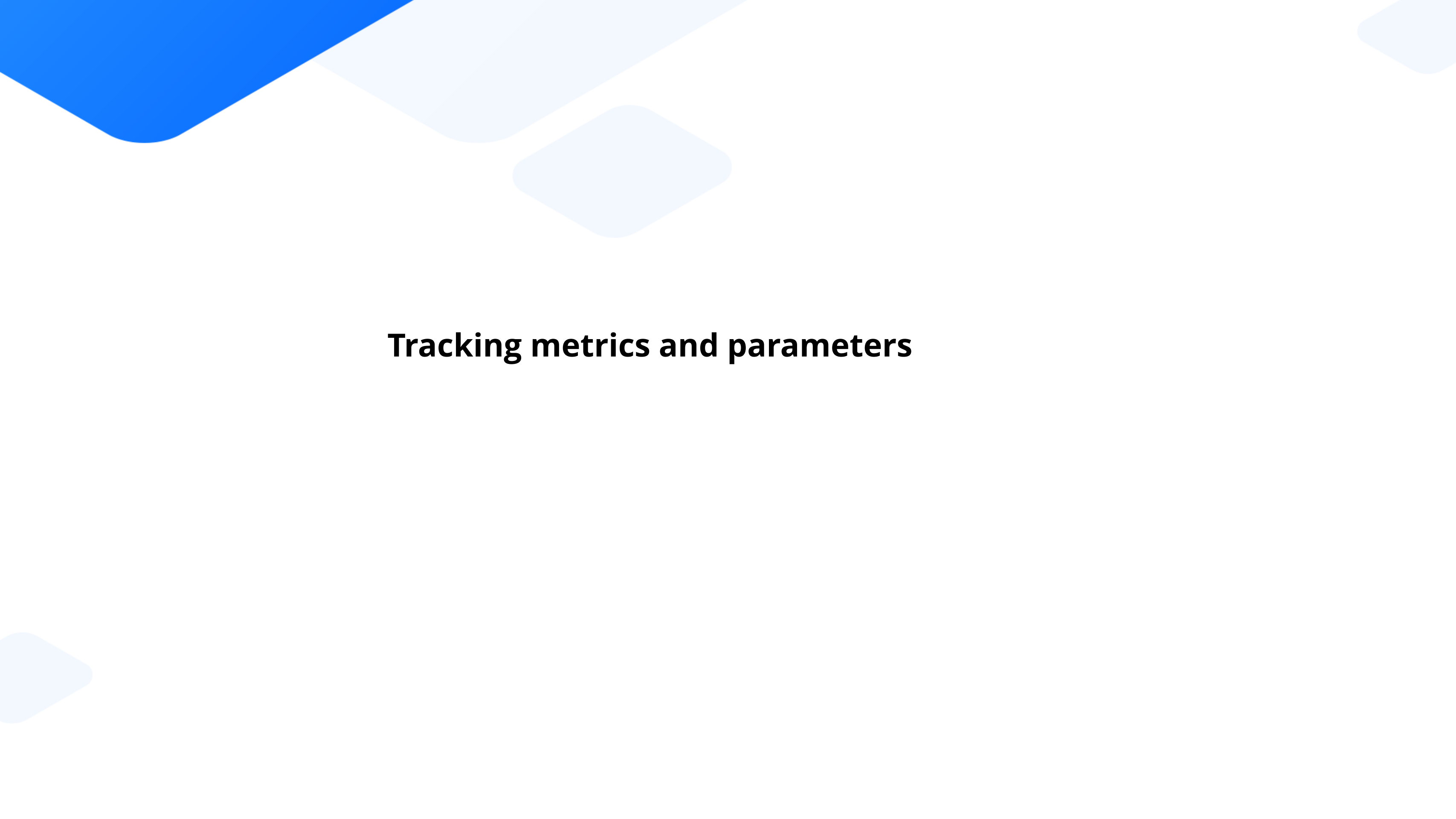
★ Experiment 3 has the highest accuracy (94%) and is selected for deployment.

4 What Should Testers Validate?

- Was a new experiment created? ✓
- Were the training parameters logged? ✓
- Was the model accuracy recorded? ✓
- Was the correct model version saved? ✓
- Can different experiments be compared? ✓
- Is the best model selected for deployment? ✓



Key Takeaway: Experiment Management helps QA teams track, compare, and select the best AI model using complete and accurate records.

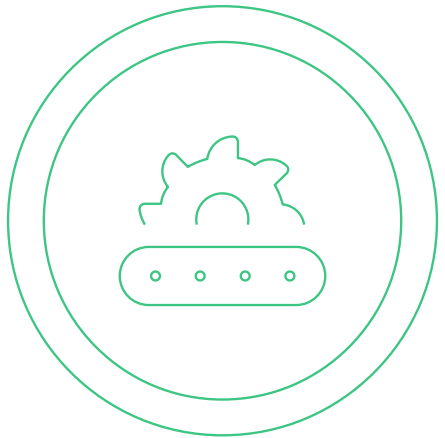


Tracking metrics and parameters

Tracking metrics and parameters

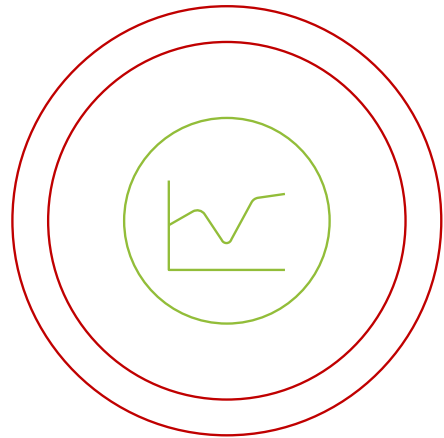
Track Model Parameters

Learning rate and iterations are tracked.



Record Performance Metrics

Accuracy, precision, recall, F1-score are recorded.



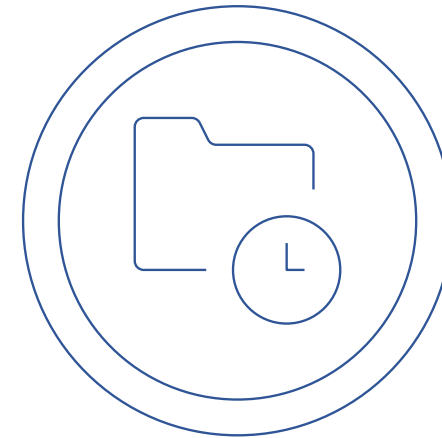
Compare Multiple Training Runs

Best-performing model is identified.



Maintain History of Every Experiment

Easy review and auditing are supported.



Support Informed Decisions

Model with best performance is selected.



Comprehensive tracking of model parameters and performance is crucial for selecting the best model and ensuring reproducibility.

VOIS EXAMPLE – TRACKING METRICS & PARAMETERS

1. TRAIN AI MODEL (3 EXPERIMENTS)



Experiment 1

LR: 0.01



Experiment 2

LR: 0.05



Experiment 3

LR: 0.10



Experiment	Learning Rate	Accuracy
Experiment 1	0.01	88%
Experiment 2	0.05	91%
Experiment 3	0.10	94%

2. TRACKED IN MLflow

mlflow



Model Parameters
(Learning Rate)



Accuracy

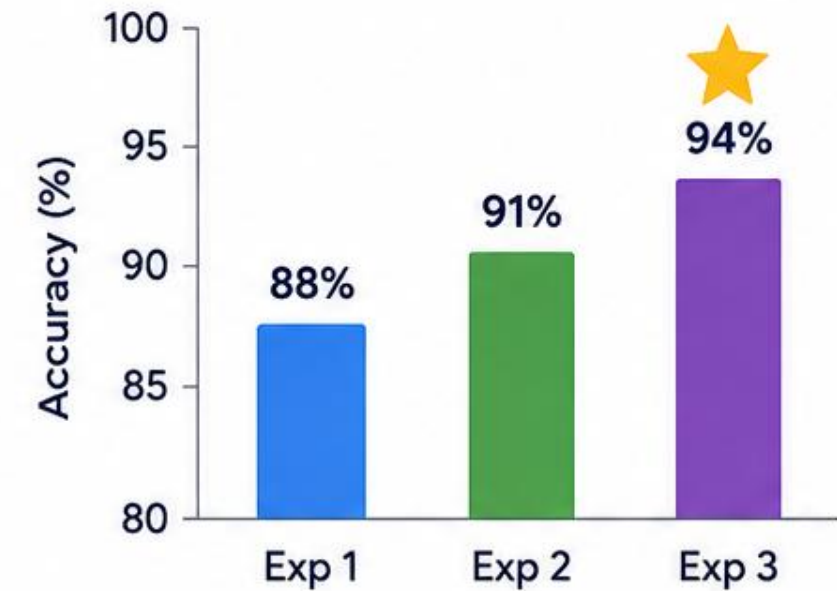


Model Version



Training Date

3. COMPARE & SELECT BEST MODEL



BEST MODEL
Experiment 3

94%

Selected for Deployment



QA Team
Trains Models



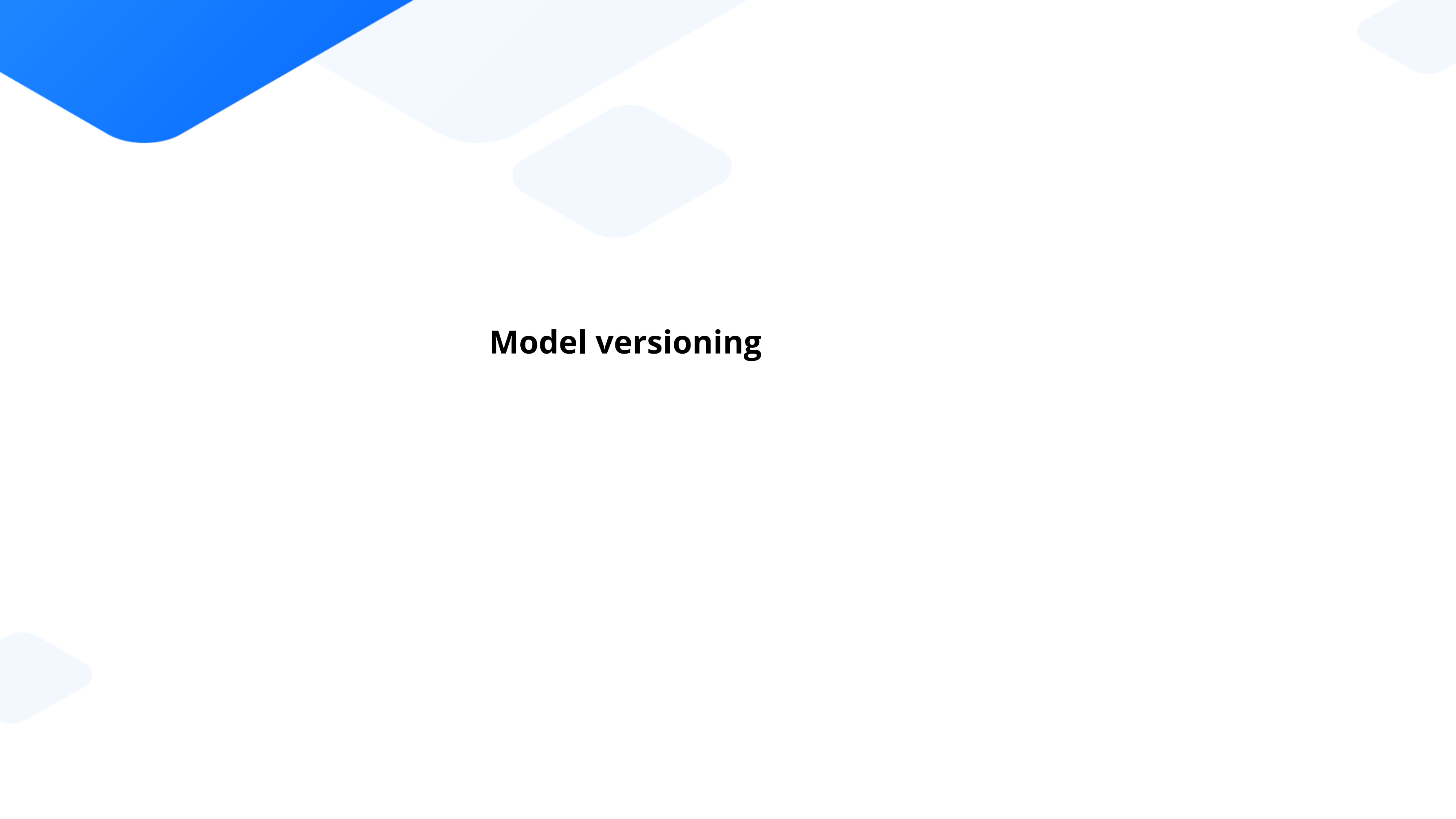
MLflow Tracks
Everything



Compare
Experiments



Deploy Best
Model



Model versioning

Model versioning

Unique Version Numbers

1 —
2 —
3 —

Assign unique numbers (v1, v2, v3) to each retrained model for easy identification.

Version Comparison



Facilitate comparison of different model versions to track improvements.

Rollback Capability



Enable rollback to a previous model if the latest version fails.

Approved Model Deployment



Ensure only approved models are deployed to production.

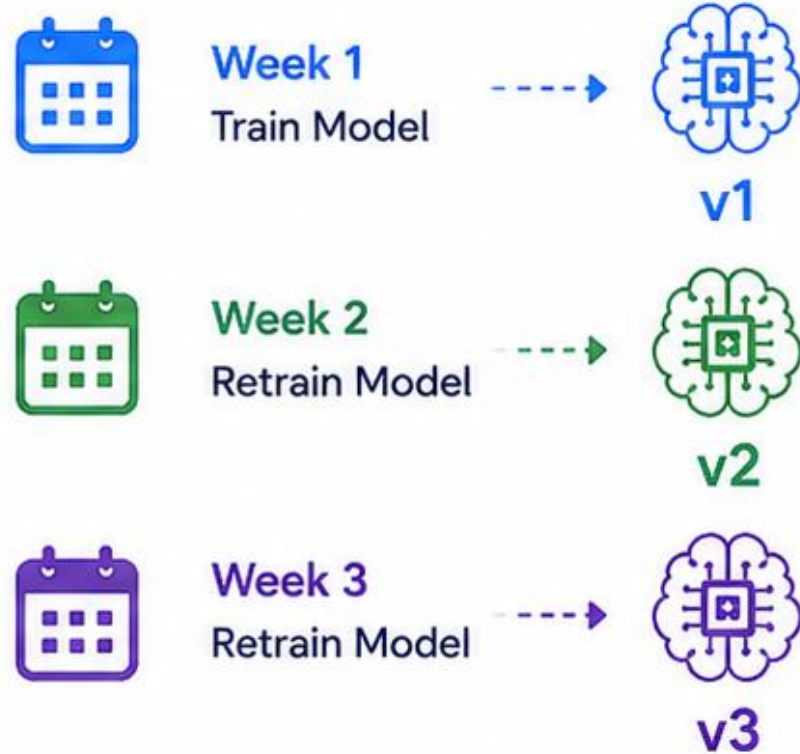


VOIS EXAMPLE – MODEL VERSIONING



The QA team retrains the AI model every week.

1. Retrain AI Model Every Week



2. Saved in MLflow Model Registry

mlflow
Model Registry



Each model is saved
as a new version

3. Compare & Deploy Best Model

Model Version	Accuracy	Status
v1	88%	● Previous Model
v2	91%	↗ Improved
v3	94%	✓ Deployed



QA Lead selects **Version 3**
because it provides the
highest accuracy.

Train Model
Every Week



QA Team

New Version
Created



Model Version

Stored in
MLflow Registry



All Versions Saved

Compare
Performance



Review Metrics

Deploy Best
Version



Version 3 Deployed

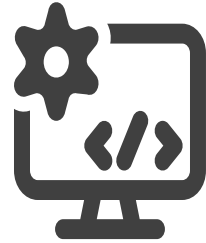
BENEFITS

- ✓ Track every model version
- ✓ Compare performance easily
- ✓ Roll back if needed
- ✓ Deploy only the best model



Model lifecycle states

Model lifecycle states



Development

The model is trained and tested by the data science team.



Staging

The model is validated by QA before production deployment.



Production

The approved model is deployed and used by end users.



Archived

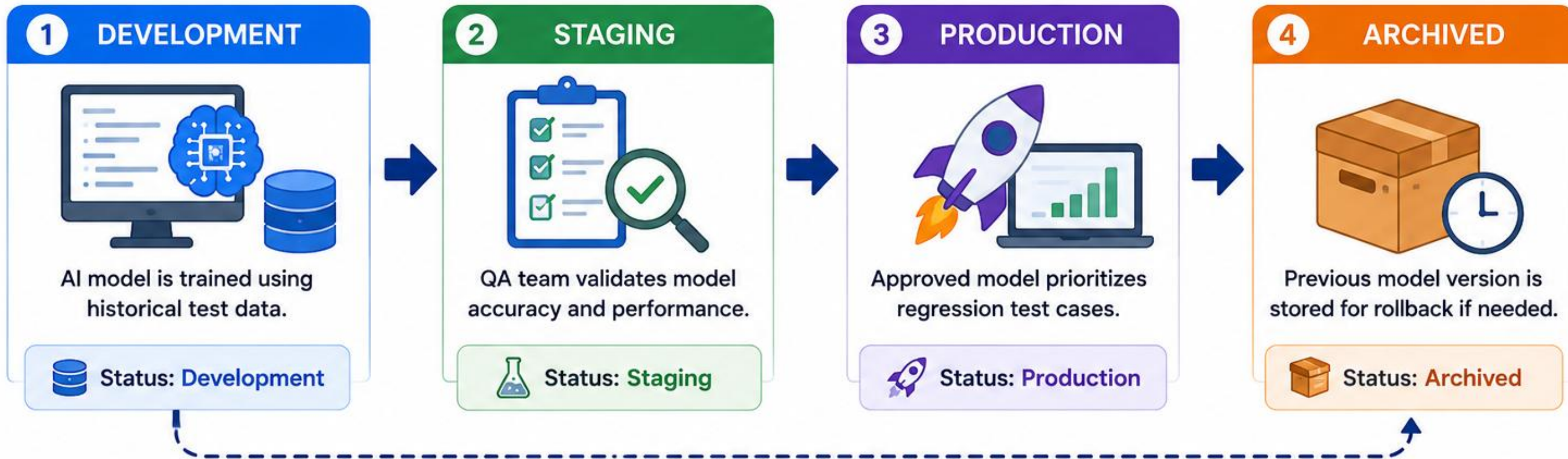
Older or unused models are stored for future reference or rollback.

Model Lifecycle States describe the journey of an AI model—from Development, to Staging, to Production, and finally to Archived—ensuring the right model is tested, deployed, and maintained throughout its lifecycle

VOIS EXAMPLE – MODEL LIFECYCLE STATES



A QA team develops a model to **prioritize regression test cases**.

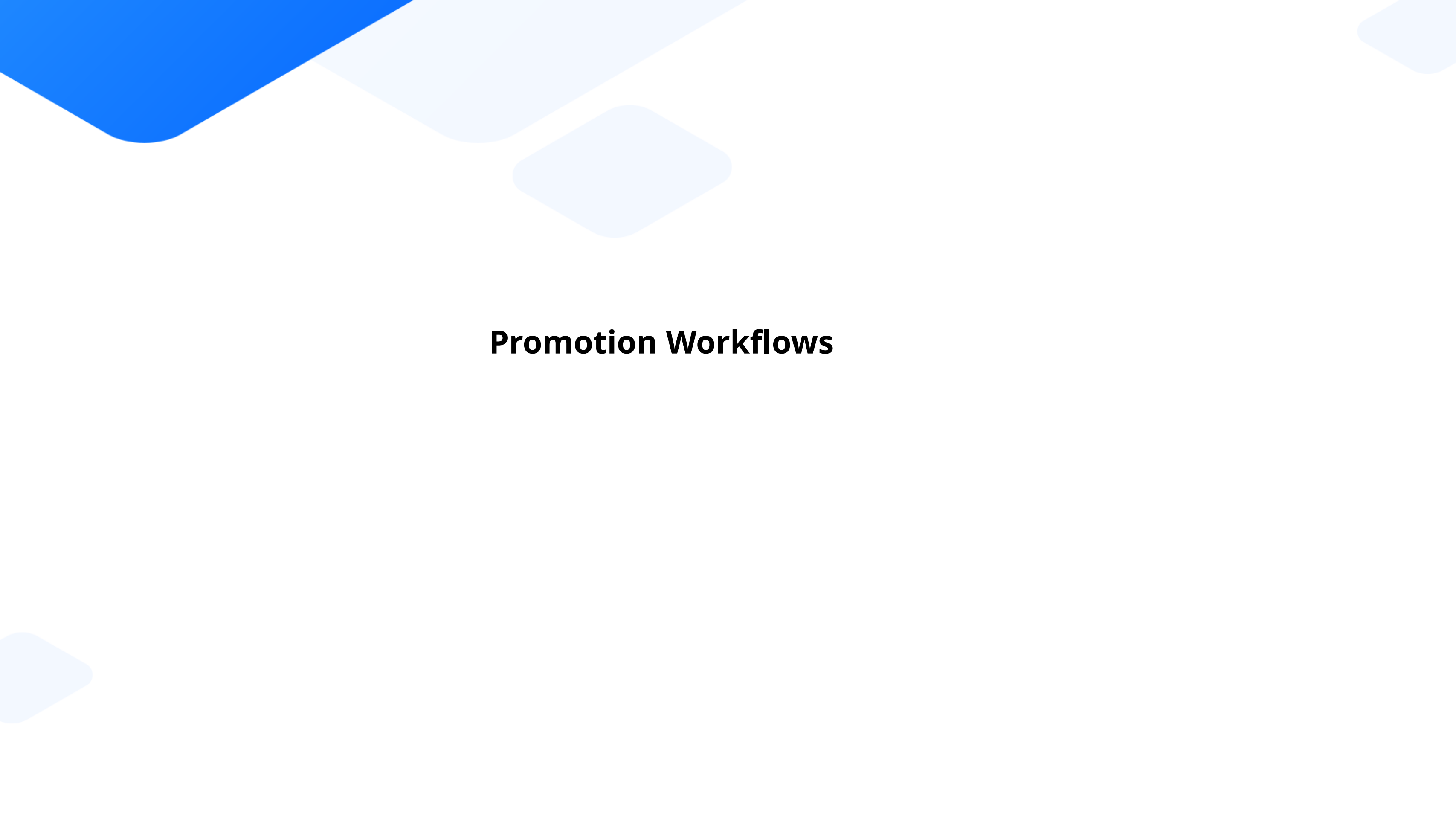


TESTER RESPONSIBILITIES

- 1** Verify the correct model moves to **Staging**.
- 2** Validate model accuracy before **Production**.
- 3** Confirm only the approved model is **deployed**.
- 4** Verify previous model versions are **archived**.
- 5** Ensure **rollback** is possible if a production issue occurs.

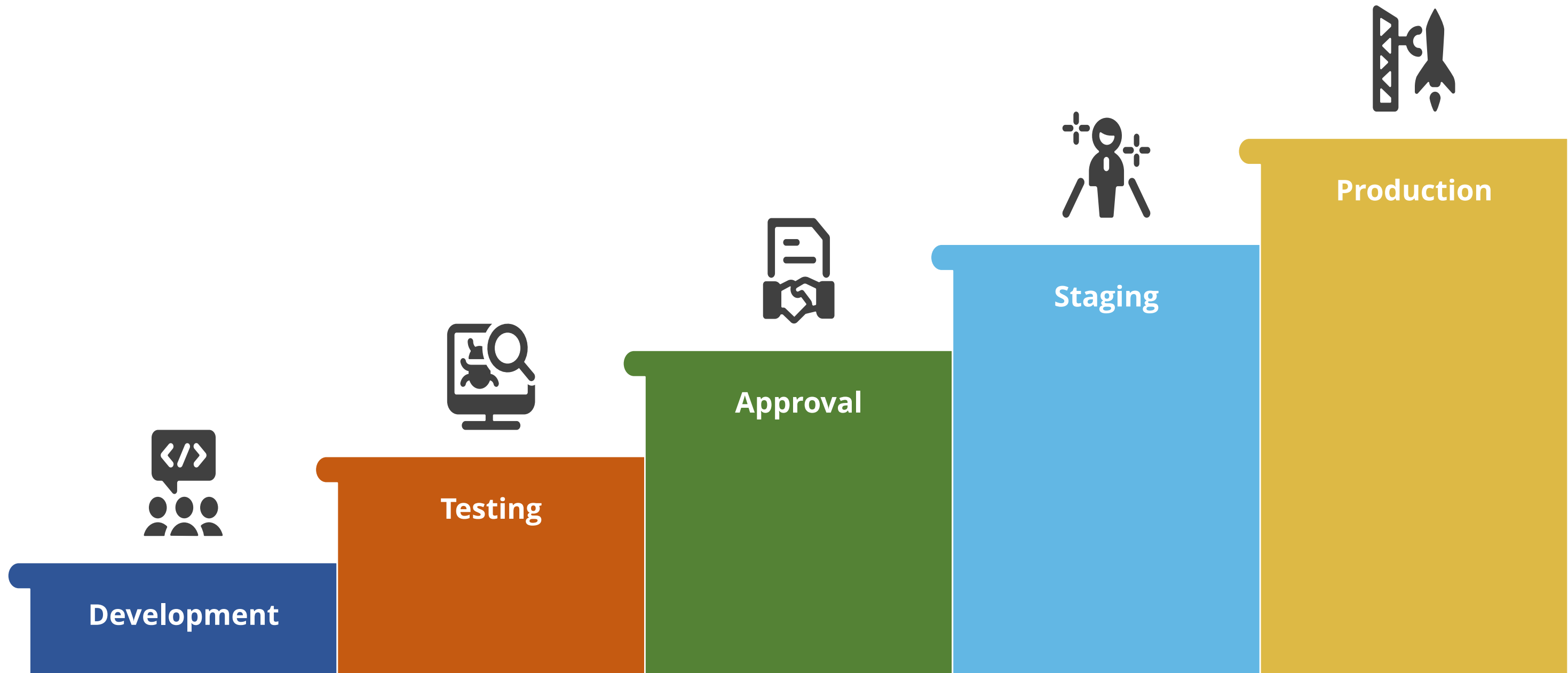


KEY TAKEAWAY: Following model lifecycle states ensures quality, control, and safety of AI models in production.



Promotion Workflows

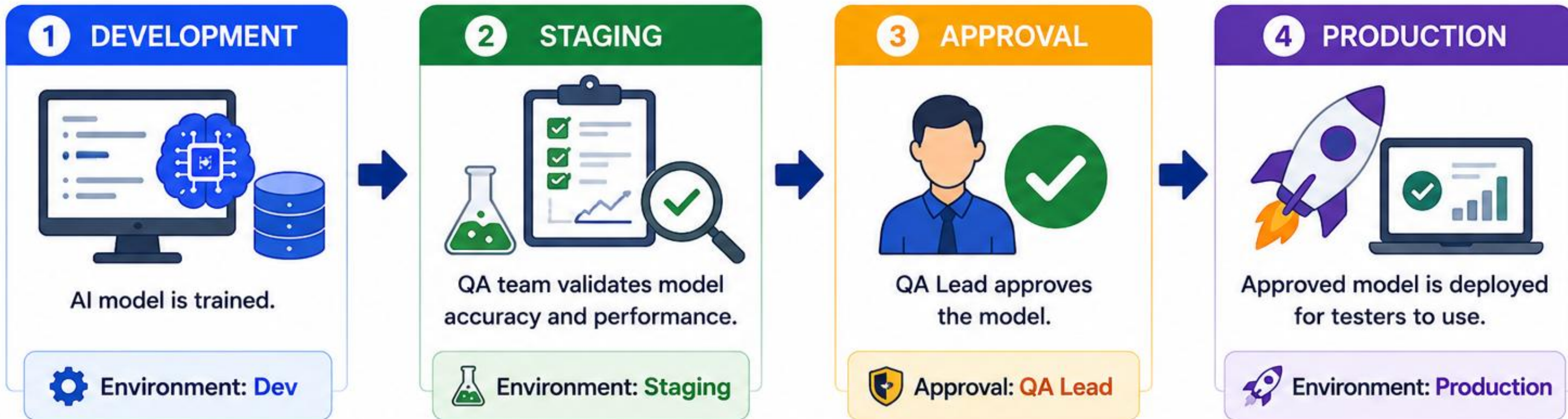
Promotion Workflows



VOIS EXAMPLE – PROMOTION WORKFLOWS



The QA team trains a new AI model to **prioritize regression test cases**.



TESTER RESPONSIBILITIES

- 1** Verify the model passed all validation tests.



- 2** Confirm the correct model version is promoted.



- 3** Ensure QA Lead approval is recorded.



- 4** Verify only the approved model reaches Production.



- 5** Validate the deployed model works correctly after promotion.



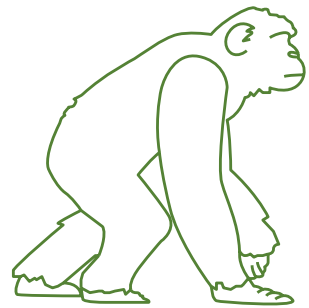
KEY TAKEAWAY: A strong promotion workflow ensures only tested, approved, and reliable AI models reach Production.





Registry integration with CI/CD

Registry integration with CI/CD



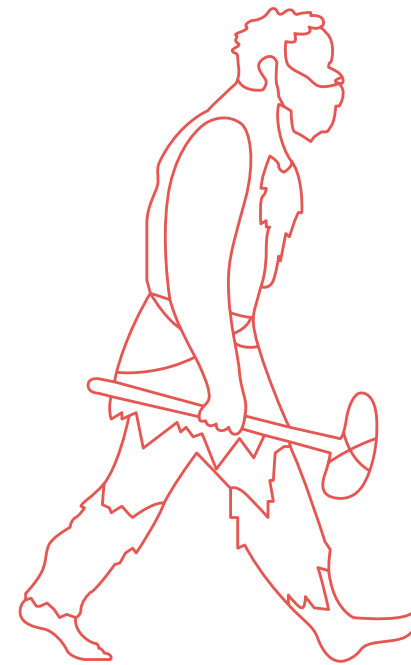
Model Registry

Approved AI models
stored



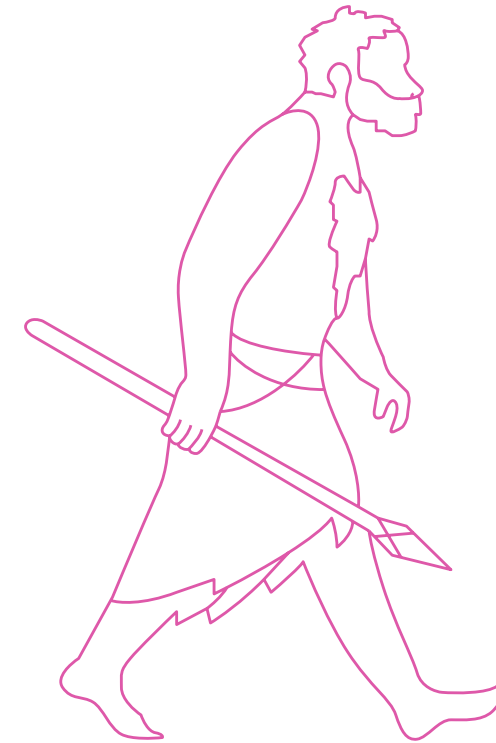
CI/CD Pipeline

Retrieves approved
model



Validation

Ensures model is
validated



Deployment

Deploys approved
model



Production

Consistent and
automated
deployments

VOIS EXAMPLE – REGISTRY INTEGRATION WITH CI/CD



VOIS develops a new software release using AI models. The Model Registry is integrated with CI/CD

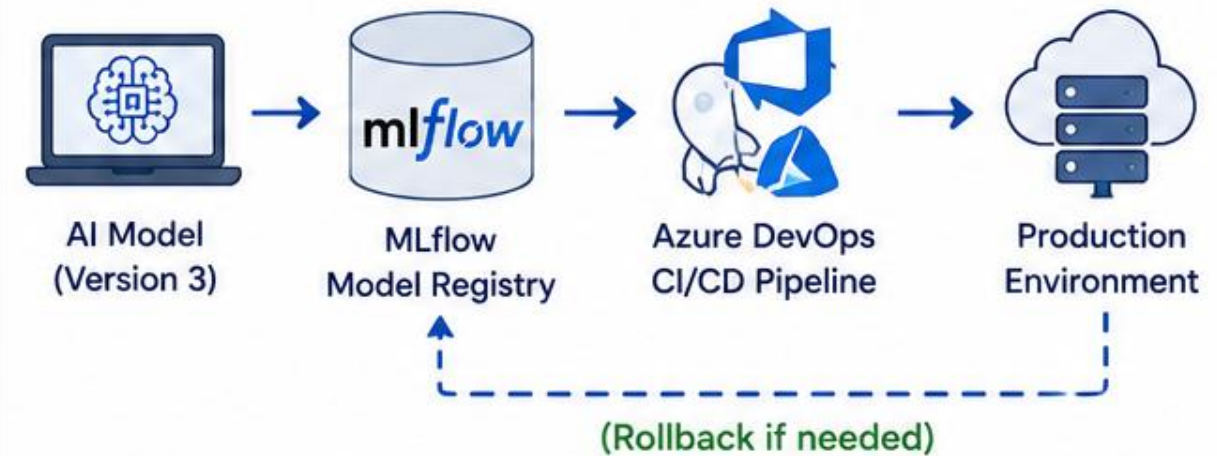
VOIS RELEASE JOURNEY

-  A new AI model (Version 3) is trained.
-  The model is registered in the **MLflow** Model Registry.
-  The QA team validates the model.
-  The QA Lead approves **Version 3**.
-  The Azure DevOps CI/CD pipeline automatically deploys **Version 3** to **Production**.
-  Testers verify that the application is using the newly deployed model.

SIMPLE WORKFLOW



INTEGRATION OVERVIEW



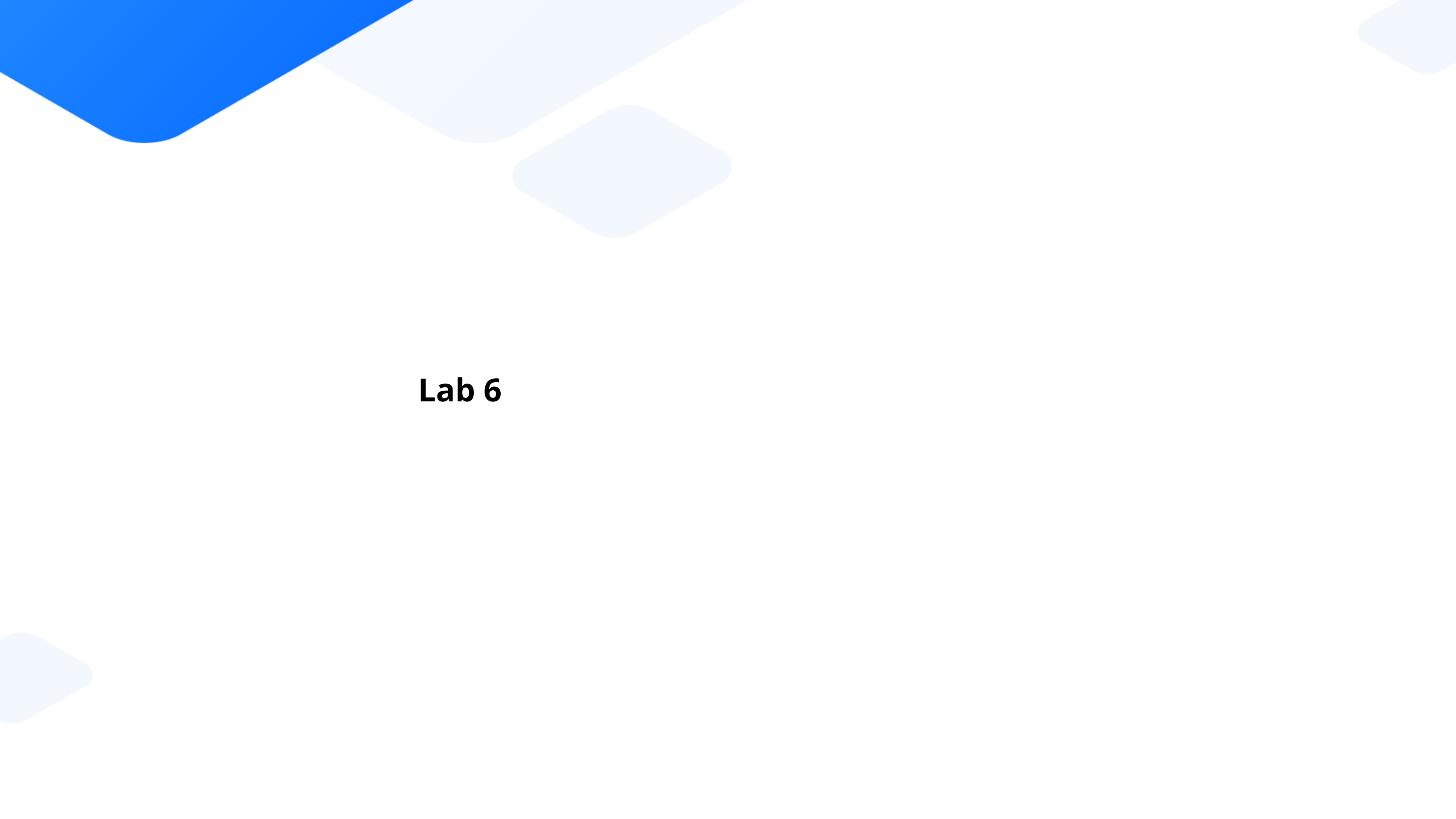
TESTER RESPONSIBILITIES

- ✓ Verify the correct model version is retrieved from the registry.
- ✓ Confirm only the approved model is deployed.
- ✓ Validate the CI/CD pipeline completed successfully.
- ✓ Verify the application uses the latest deployed model.
- ✓ Ensure rollback is possible if deployment fails.



KEY TAKEAWAY: Integrating Model Registry with CI/CD ensures only tested, approved AI models are automatically deployed to Production – faster, safer, and reliable.





Lab 6

QNA